

VoxTrust-3D: Scribble-Calibrated Risk-Aware Pseudo-Label Transfer for 3D Medical Image Segmentation

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Abstract

Scribble supervision reduces annotation cost but provides direct labels only on a small subset of voxels. The main difficulty is therefore not only how to generate dense pseudo-labels, but how to decide which transferred labels are reliable enough to train on. We propose VOXTRUST-3D, a simple direct-3D framework that places the main contribution in reliability control rather than network complexity. A single 3D U-Net student is paired with an exponential-moving-average teacher. The teacher proposes soft voxel-wise targets, while a reliability score combines semantic margin with weak-to-strong perturbation stability. The key component is a spatially blocked scribble calibrator: a small subset of complete scribble blocks is reserved from direct supervision and used only to measure whether high-reliability teacher predictions are actually correct. Reliability thresholds are estimated separately for the teacher-predicted class and the physical 3D distance from observed supervision. A conservative lower-confidence rule selects pseudo-labels and explicitly abstains when calibration evidence is insufficient. Training and inference remain fully volumetric; no slice-wise 2D training or post-hoc slice stacking is used. We also define mechanism-level diagnostics and ablations that can directly test whether the proposed calibration improves pseudo-label precision at matched coverage, instead of relying only on final Dice scores.

1. Motivation

Dense voxel labels are expensive for 3D medical images. Scribbles reduce this cost, but they change the learning problem: labeled voxels are sparse, while the remaining voxels are *unknown*, not background. A model must therefore transfer class information from a small observed set to a much larger unknown region without amplifying its own errors.

Recent studies improve this transfer with soft pseudo-labels, multi-branch uncertainty, consistency, feature prototypes, or distance priors [1–5]. However, most reliability rules are derived from the model itself. High confidence, low entropy, or agreement between correlated predictors can

still be wrong. ScribbleBench further shows that several specialized designs do not generalize well across tasks, while a 3D nnU-Net with partial loss is a strong baseline [6]. This motivates a simple architecture with a stronger training rule.

Our core idea is to use part of the *existing scribbles* as calibration evidence. If a reliability score is meaningful, high-scoring predictions on spatially held-out scribble regions should be correct. In a volume, this correctness may depend on both semantic class and how far supervision is transferred in physical space. VOXTRUST-3D therefore estimates reliability thresholds conditioned on class and physical transfer distance, and it allows the model to abstain when the evidence is weak.

Contributions. We make four method contributions:

- (1) a direct-3D teacher-student framework in which pseudo-label selection, rather than architecture complexity, is the main learning mechanism;
- (2) a spatially blocked scribble partition that separates direct fitting from reliability calibration and reduces local calibration leakage;
- (3) class- and physical-distance-conditioned reliability calibration with conservative lower-confidence selection and explicit abstention;
- (4) a reliability-centered evaluation plan that measures pseudo-label precision, coverage, calibration, and performance versus distance from the nearest scribble, in addition to segmentation metrics.

2. Positioning Against Related Work

Pseudo-label reliability is already central to scribble-supervised segmentation. DMSPS uses dynamically mixed soft labels and uncertainty-based annotation expansion [1]. MaCo creates continuous labels from distance maps [2]. PS-Seg performs direct 3D learning with three decoders and voxel-wise uncertainty [3], while EDHAF uses epistemic hardness to focus training on difficult regions [4]. FW2SS combines weak-to-strong consistency with prototype contrastive calibration [5]. In semi-supervised learning, URPC uses disagreement across prediction scales as an efficient

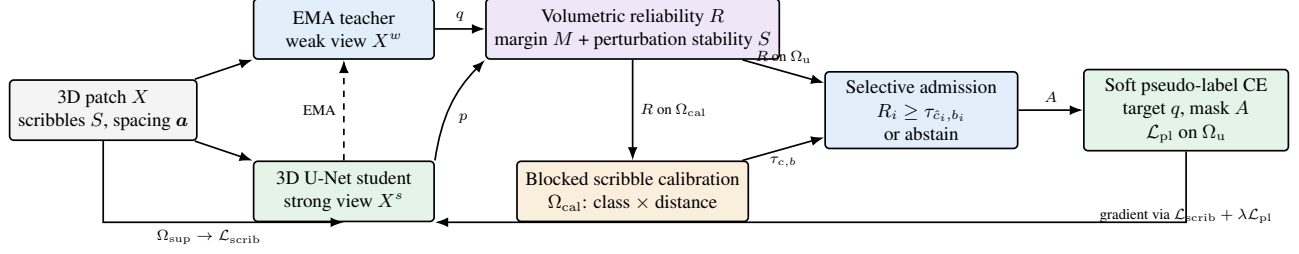


Figure 1. Training overview of VOXTRUST-3D. Complete scribble blocks are split once into direct-supervision voxels Ω_{sup} and calibration voxels Ω_{cal} . The teacher proposes soft targets; the student supplies a perturbation view. Reliability is calibrated against held-out scribble blocks by predicted class and physical 3D transfer distance. Only accepted unknown voxels receive soft teacher supervision. The calibrator is removed at inference.

Table 1. Method-level distinction from close prior work. “Calib.” means that observed labels are explicitly used to calibrate pseudo-label acceptance.

Method	Direct 3D	Soft PL	Calib.	Dist.-cond.
DMSPS [1]	✓	✓	✗	✗
MaCo [2]	✗	✓	✗	fixed prior
PS-Seg [3]	✓	✓	✗	✗
EDHAF [4]	✓	✓	✗	✗
FW2SS [5]	mostly 2D	✓	✗	✗
VOXTRUST-3D	✓	✓	✓	✓

uncertainty signal [7], and UniMatch shows that weak-to-strong consistency is a strong basic recipe [8].

Conformal pseudo-label selection has also been explored in semi-supervised medical segmentation [9]. That line of work is important because it separates confidence from calibrated selection. However, standard conformal guarantees require exchangeability assumptions that are difficult to justify for highly correlated scribble voxels. We therefore do *not* claim a conformal guarantee. Instead, we use a spatial hold-out and a conservative empirical lower bound as a practical risk-control rule.

3. Problem Formulation

Let the training set be $\mathcal{D} = \{(X^{(n)}, S^{(n)}, \mathbf{a}^{(n)})\}_{n=1}^N$, where X is a 3D image, S is a scribble map, and $\mathbf{a} = (a_x, a_y, a_z)$ is voxel spacing in millimeters. Let Ω denote the voxel domain. Scribble labels are observed only on $\Omega_s \subset \Omega$; all remaining voxels form $\Omega_u = \Omega \setminus \Omega_s$. For C classes, the student f_θ and EMA teacher $f_{\bar{\theta}}$ predict

$$p_i = \text{softmax}(f_\theta(X^s))_i, \quad q_i = \text{softmax}(f_{\bar{\theta}}(X^w))_i, \quad (1)$$

where X^w and X^s share exactly the same spatial transform. The strong view receives additional intensity perturbations only, so p_i , q_i , and S_i remain voxel-aligned.

The teacher is updated after each student step using Mean Teacher [10]:

$$\bar{\theta}_t = \alpha \bar{\theta}_{t-1} + (1 - \alpha) \theta_t. \quad (2)$$

The teacher is a stable target generator, not an independent source of truth; therefore teacher-student agreement is used only as one cue of reliability.

4. Proposed Method

4.1. Spatially Blocked Scribble Calibration

We split the observed scribbles once at the beginning of a training run:

$$\Omega_s = \Omega_{\text{sup}} \cup \Omega_{\text{cal}}, \quad \Omega_{\text{sup}} \cap \Omega_{\text{cal}} = \emptyset. \quad (3)$$

Ω_{sup} is used for gradient-based supervised learning. Ω_{cal} is used only to test whether a reliability score predicts correctness. Calibration voxels never enter the supervised or pseudo-label loss.

The split is made at a *spatial block* level rather than by independent voxels. For volumetric data, the default block is all scribble voxels of one class on one annotated slice; for a true 3D stroke, a connected stroke component is used instead. Thus a held-out class-slice block is separated from direct supervision on that class in the same plane. This reduces the leakage that occurs when calibration voxels are interleaved with supervised voxels on the same stroke. We reserve a fraction η of available blocks per class, while keeping at least one supervised block whenever possible. A practical initial value is $\eta = 0.15$; the exact value is validated once and then fixed. No additional annotation is introduced: the held-out blocks come from the original scribble budget.

Direct supervision is partial cross-entropy:

$$\mathcal{L}_{\text{scrib}} = -\frac{1}{|\Omega_{\text{sup}}|} \sum_{i \in \Omega_{\text{sup}}} \sum_{c=1}^C y_{ic} \log p_{ic}. \quad (4)$$

No loss is applied to Ω_u before pseudo-label selection, and unlabeled voxels are never interpreted as background.

4.2. Volumetric Reliability Score

For each candidate voxel, the teacher predicts $\hat{c}_i = \arg \max_c q_{ic}$. We use three low-cost cues that capture different failure modes.

Semantic margin.

$$M_i = q_{i\hat{c}_i} - \max_{c \neq \hat{c}_i} q_{ic}. \quad (5)$$

Unlike maximum confidence alone, the margin also measures how competitive the second-best class is.

Weak-to-strong stability. We measure prediction stability under appearance perturbation with Jensen-Shannon divergence:

$$\begin{aligned} S_i &= 1 - \frac{D_{\text{JS}}(p_i, q_i)}{\log 2}, \\ m &= \frac{1}{2}(p + q), \\ D_{\text{JS}}(p, q) &= \frac{1}{2}D_{\text{KL}}(p \| m) + \frac{1}{2}D_{\text{KL}}(q \| m). \end{aligned} \quad (6)$$

This score lies in $[0, 1]$. Agreement is useful, but not sufficient, because teacher and student errors can be correlated.

Combined score. The raw reliability score is the geometric mean

$$R_i = \sqrt{\text{clip}(M_i, \epsilon, 1) \text{clip}(S_i, \epsilon, 1)}. \quad (7)$$

R_i is *not* interpreted as a calibrated probability. It only ranks candidate pseudo-labels; the held-out scribbles determine which score values are trusted. We deliberately keep the core score to two inexpensive cues. Cross-scale disagreement, motivated by URPC [7], is evaluated only as an optional ablation rather than required by the framework.

4.3. Physical 3D Transfer Distance

For every class c present in Ω_{sup} , we precompute a physical Euclidean distance map

$$D_c(i) = \min_{j \in \Omega_{\text{sup}}^{(c)}} \|\text{diag}(\mathbf{a})(\mathbf{v}_i - \mathbf{v}_j)\|_2, \quad (8)$$

where $\mathbf{v}_i$ is the integer voxel coordinate and $\Omega_{\text{sup}}^{(c)}$ contains supervised scribble voxels of class c . Distances are measured in millimeters, so in-plane and through-plane spacing are handled consistently.

For a candidate predicted as class $\hat{c}_i$, its transfer distance is $d_i = D_{\hat{c}_i}(i)$. We use B distance strata, with $B = 3$ (near, middle, far) by default. Before optimization, class-specific bin edges are computed from the physical distances of the held-out calibration scribbles to Ω_{sup} and pooled across training volumes; quantile edges give roughly balanced calibration support without using dense masks. We also store the largest supported calibration distance d_c^{max} for each class. A candidate with $d_i > d_{\hat{c}_i}^{\text{max}}$ is outside the observed calibration support and is rejected rather than extrapolated. Importantly, the method does not impose a monotonic distance-decay

prior: distance defines where reliability evidence is compared, not how confidence must decay.

If a predicted class has no supervised scribble in the parent volume, a distance-conditioned threshold is undefined. The method first falls back to a class-only threshold pooled across distances; if class-level calibration is also insufficient, it abstains. For rare classes with too few held-out distances to define class-specific bin edges, global calibration-distance quantiles are used only to define the strata; the acceptance threshold remains class-conditioned.

4.4. Class- and Distance-Conditioned Risk Calibration

For each calibration voxel $j \in \Omega_{\text{cal}}$, we record four quantities: reliability R_j , teacher prediction $\hat{c}_j$, distance stratum b_j , and correctness

$$e_j = \mathbb{I}[\hat{c}_j = y_j]. \quad (9)$$

Thresholds are conditioned on the *teacher-predicted* class, because the same predicted class determines which threshold is applied to an unknown voxel.

To keep the estimator responsive during training without assigning fractional sample counts, each pair (c, b) maintains a rolling memory

$$\mathcal{M}_{c,b} = \{(R_j, e_j)\}_{j=1}^{N_{c,b}}, \quad N_{c,b} \leq N_{\text{max}}. \quad (10)$$

Each spatial calibration block contributes at most m_{max} uniformly sampled voxels per update so that long strokes do not dominate the memory. When the buffer is full, the oldest observations are removed. A class-only buffer $\mathcal{M}_c^{\text{cls}}$ is maintained in parallel by pooling calibration observations across valid distance strata. If a calibration prediction has no valid class-specific distance in its volume, it updates only $\mathcal{M}_c^{\text{cls}}$.

For a candidate score threshold t , define

$$n_{c,b}(t) = \sum_{(r,e) \in \mathcal{M}_{c,b}} \mathbb{I}[r \geq t], \quad (11)$$

$$k_{c,b}(t) = \sum_{(r,e) \in \mathcal{M}_{c,b}} \mathbb{I}[r \geq t]e, \quad (12)$$

$$\hat{\pi}_{c,b}(t) = \frac{k_{c,b}(t)}{\max(n_{c,b}(t), 1)}. \quad (13)$$

Instead of trusting raw empirical precision, we use the Wilson lower confidence bound

$$\text{LCB}(\hat{\pi}, n) = \frac{\hat{\pi} + \frac{z^2}{2n} - z \sqrt{\frac{\hat{\pi}(1-\hat{\pi})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \quad (14)$$

where $z = \Phi^{-1}(1 - \delta/2)$. The accepted threshold is

$$\tau_{c,b} = \min\{t \in \mathcal{T} : n_{c,b}(t) \geq n_{\text{min}}, \text{LCB}(\hat{\pi}_{c,b}(t), n_{c,b}(t)) \geq \rho\}. \quad (15)$$

$\mathcal{T}$ is a fixed score grid, e.g. $\{0, 0.01, \dots, 1\}$. If no threshold satisfies Eq. (15), $\tau_{c,b} = +\infty$ and that stratum abstains. The class-only fallback threshold τ_c^{cls} is computed by the same rule after replacing $\mathcal{M}_{c,b}$ with $\mathcal{M}_c^{\text{cls}}$. We use the Wilson bound as a conservative practical rule, *not* as a formal coverage guarantee, because medical scribble voxels remain spatially correlated.

4.5. Selective Soft Pseudo-Label Learning

For an unknown voxel $i \in \Omega_u$, acceptance is

$$A_i = \mathbb{I}[i \in \Omega_u] \mathbb{I}[d_i \leq d_{\hat{c}_i}^{\max}] \mathbb{I}[R_i \geq \tau_{\hat{c}_i, b_i}]. \quad (16)$$

When d_i is undefined because class $\hat{c}_i$ has no supervised scribble in the current volume, the distance-support term in Eq. (16) is replaced by the class-only acceptance rule $R_i \geq \tau_{\hat{c}_i}^{\text{cls}}$. If that threshold is unavailable, $A_i = 0$. The target is the detached teacher distribution $\text{sg}(q_i)$, not a hard one-hot label. The pseudo-label loss is

$$\mathcal{L}_{\text{pl}} = - \frac{1}{\sum_i A_i + \epsilon} \sum_{i \in \Omega_u} A_i \sum_{c=1}^C \text{sg}(q_{ic}) \log p_{ic}. \quad (17)$$

Using soft targets keeps residual class ambiguity and avoids adding another handcrafted weighting module. Reliability-weighted soft CE can be evaluated as an ablation, but is not part of the core method.

The complete objective is

$$\mathcal{L}(t) = \mathcal{L}_{\text{scrib}} + \lambda(t) \mathcal{L}_{\text{pl}}, \quad (18)$$

where $\lambda(t)$ is ramped from zero after a scribble-only warm-up. Calibration memories are initialized empty at the end of the warm-up, so early untrained predictions do not affect later thresholds. Pseudo-label learning is activated for a stratum only after its calibration memory has at least $n_{\min}$ samples. This creates a natural curriculum: well-supported regions enter training early; uncertain regions remain unlabeled. All reliability scores, thresholds, and admission masks are stop-gradient quantities; gradients flow only through the segmentation losses.

5. Training and Inference Details

Direct 3D optimization. Training uses tensors of shape $[B, C, D, H, W]$ and 3D patches, not independent slices. Patch-based optimization is still fully volumetric because every prediction uses through-plane context. We recommend a standard 3D U-Net or nnU-Net configuration [11, 12]. Inference uses 3D sliding windows with overlap and Gaussian blending to reconstruct the full probability volume. Only one EMA network is required at inference; the calibrator is removed. If an nnU-Net backbone is used, its standard deep-supervision configuration can be kept identical for all compared methods, but it is not required by our reliability score.

Algorithm 1 One training iteration of VOXTRUST-3D

Require: 3D patch (X, S) , spacing $\mathbf{a}$, student f_θ , EMA teacher $f_{\bar{\theta}}$, calibration memories

- 1: Apply one common spatial transform to image and scribbles
 - 2: Build weak view X^w and strong intensity view X^s
 - 3: $q \leftarrow \text{softmax}(f_{\bar{\theta}}(X^w))$ without gradients
 - 4: $p \leftarrow \text{softmax}(f_\theta(X^s))$
 - 5: Compute $\mathcal{L}_{\text{scrib}}$ on Ω_{sup} by Eq. (4)
 - 6: $\mathcal{L}_{\text{pl}} \leftarrow 0$
 - 7: **if** scribble-only warm-up has finished **then**
 - 8: Compute detached $R_j, \hat{c}_j, b_j, e_j$ for $j \in \Omega_{\text{cal}}$
 - 9: Update $\mathcal{M}_{\hat{c}_j, b_j}$ and class-only $\mathcal{M}_{\hat{c}_j}^{\text{cls}}$
 - 10: Fit $\tau_{c,b}$ by Eq. (15); use class-only fallback only when distance is undefined
 - 11: Compute detached $R_i, \hat{c}_i, b_i$ for $i \in \Omega_u$
 - 12: Reject out-of-support distances; build A_i by Eq. (16) and compute $\mathcal{L}_{\text{pl}}$
 - 13: **end if**
 - 14: Update θ using $\mathcal{L}(t)$ in Eq. (18)
 - 15: Update $\bar{\theta} \leftarrow \alpha \bar{\theta} + (1 - \alpha) \theta$
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Spacing-aware preprocessing. ACDC and MS-CMRSeg contain anisotropic cardiac MRI, while WORD contains abdominal CT with a different spacing profile [13–15]. Preprocessing should therefore follow dataset-specific nnU-Net spacing rules rather than forcing all data to an arbitrary isotropic grid. The distance maps in Eq. (8) use the spacing after the same resampling applied to training data.

Augmentation. Spatial augmentations (rotation, scaling, mirroring, elastic deformation if used) are sampled once and applied identically to image and scribble coordinates. After this common transform, the teacher receives the weak view and the student receives additional brightness, contrast, gamma, Gaussian noise, or mild blur. This follows the weak-to-strong idea [8] while preserving voxel alignment required by the calibrator. Cutout and copy-paste are excluded from the core method because they can erase the already sparse scribble evidence.

Recommended initial hyperparameters. A practical starting configuration is $\alpha = 0.99$ for the EMA teacher, $\eta = 0.15$ held-out scribble blocks, $B = 3$ distance strata, target precision $\rho = 0.95$, $\delta = 0.05$, $n_{\min} = 32$, $N_{\max} = 4096$ observations per stratum, $m_{\max} = 64$ sampled voxels per calibration block per update, and a pseudo-label ramp-up over the first 20%–30% of training. These are starting values, not dataset-specific claims; they must be tuned on validation data only.

Edge cases. If a class has only one scribble block in a volume, that block remains in Ω_{sup} and calibration evidence is collected from other volumes. If a class-distance stratum

Table 2. Each component must answer one failure mode and has a removal criterion.

Component	Purpose	Remove if...
EMA teacher	reduce target noise	no gain over current-model targets
Margin M	detect class ambiguity	no better ranking than max confidence
Perturbation S	detect unstable predictions	no precision gain at matched coverage
Optional scale cue	test whether cross-scale disagreement adds signal	no gain beyond M and S
Blocked calibration	test score against unseen scribble regions	random-pixel hold-out is equivalent
Class $\times$ distance threshold	model heterogeneous transfer risk	no gain under sparse-slice stress tests
LCB rule	avoid small-sample overconfidence	causes abstention without reliability gain

lacks enough calibration observations, the method falls back to a class-only memory. If class-level evidence is also insufficient, it abstains. This hierarchy prevents the method from inventing supervision only to maintain coverage.

6. Why the Design Is Minimal

The framework deliberately avoids adding a second student, a transformer branch, a contrastive loss, a shape prior, or a foundation model. These components can be effective, but they would make it harder to isolate whether improvement comes from the proposed reliability mechanism. The role of each retained component is explicit:

This design also follows the lesson from ScribbleBench: new modules should be justified under diverse 3D tasks and compared to a strong partial-loss baseline [6].

7. Verification Protocol

The central claim is about *reliable supervision transfer*, so final Dice alone is insufficient. We propose evaluation on ACDC [13], MS-CMRSeg [14], and WORD [15], using matched 3D preprocessing and fixed scribble protocols whenever possible.

Primary metrics. Report Dice, HD95, and Surface Dice for segmentation. For the mechanism itself, use the dense reference masks *only for evaluation* to measure accepted pseudo-label precision, accepted coverage, class-wise precision/coverage, and risk-coverage curves. Also report pseudo-label error and final segmentation error as functions of physical distance to the nearest supervised scribble. This directly tests the motivation for distance-conditioned calibration.

Ablations. The minimum ablation sequence is: (A0) 3D partial CE; (A1) + EMA soft pseudo-labels; (A2) + fixed global threshold; (A3) + raw R_i with a fixed threshold; (A4) + class-only calibration; (A5) + spatially blocked hold-out;

(A6) + class $\times$ distance calibration; (A7) + optional multi-scale stability; and (A8) + Wilson lower-bound selection. Every added module should improve either segmentation or pseudo-label reliability at matched coverage. Otherwise it should be removed.

Sparse-slice stress test. Retain scribbles on 100%, 50%, 25%, 12.5%, and a very small subset of slices, while leaving the dense test masks unchanged. Measure performance versus millimeter distance to the nearest annotated slice. The key question is whether VOXTRUST-3D becomes more conservative as evidence becomes sparse and degrades more slowly than fixed-threshold self-training.

Falsifiable criteria. The proposed mechanism is not supported if (i) it fails to improve over the 3D partial-loss baseline on at least two distinct task types, (ii) Dice improves but pseudo-label precision at matched coverage does not, (iii) gains appear only on cardiac MRI and not on WORD, or (iv) distance conditioning gives no benefit when slice coverage is reduced. These criteria are intentionally strict to prevent over-engineering.

8. Discussion

The proposed method separates two questions that are often mixed together: *how to propose a dense target* and *whether that target should be trusted*. The EMA teacher answers the first question. The scribble calibrator answers the second using labels that already exist in the annotation budget. Conditioning calibration on physical distance gives the 3D setting a direct methodological role rather than using a 3D backbone only as an implementation detail.

The main limitation is that calibration scribbles are still selected by the annotation process and may not represent the full unknown-region distribution, especially at difficult boundaries or omitted components. Spatial blocking reduces local leakage but does not remove this selection bias. For this reason, VOXTRUST-3D uses abstention and conservative thresholding, and does not claim a formal coverage guarantee. This limitation should be measured explicitly by analyzing reliability as a function of boundary distance, class size, and slice coverage.

9. Conclusion

VOXTRUST-3D is a direct-3D scribble-supervised training framework built around a simple principle: pseudo-labels should be accepted only when their reliability is supported by observed scribble evidence. A single 3D U-Net and EMA teacher generate volumetric predictions; a spatially blocked subset of scribbles calibrates reliability by class and physical transfer distance; and only accepted unknown voxels receive soft teacher supervision. The framework is intentionally

small so that its central claim can be tested directly through precision-coverage behavior and controlled 3D ablations.

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